SHREEJEET SAHAY

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EDUCATION

University of Virginia, Charlottesville, VA — ***Aug 2024 – Dec 2026***
MS in Computer Science (GPA: 3.885/4.0)

University of Pune, Pune, India — ***Jul 2015 – Jul 2019***
BE in Information Technology (CGPA: 9.28/10.0)

PROJECTS

Thesis: Attributed Graph Alignment For Circuit Distillation — ***Sep 2025 – Present***

- Reformulated circuit distillation head mapping as attributed graph alignment, replacing ablation heuristic for the same
- Improved circuit accuracies by +0.106 for entity tracking, and +0.05 for Theory of Mind tasks over the ablation impact baseline
- Cut mapping computation time ~60x using gradient-based attribution patching instead of per-head ablation sweeps
- Reduced mapping degeneracy by replacing greedy matching with transport-based alignment
- Github: https://github.com/shreejeetsahay/circuit-distillation-attributed-network

Vision Transformer Interpretability: Assessing Attention Rollout — ***Mar 2025 – Apr 2025***

- Evaluated the efficacy of attention rollout mechanism for post-hoc interpretability in vision transformers
- Performed single-object, multi-object and embedding analyses on DeiT-Small, ViT-Base, and DETR models
- Improved object-localization saliency by 16–18% and cut inter-class embedding similarity by 40% (0.541 to 0.326)
- Identified a 12% attention entropy increase in multi-object scenes, highlighting an oversmoothing limitation in complex scenes
- Github: https://github.com/shreejeetsahay/attention-rollout-experiments

WORK EXPERIENCE

Data Science and Engineering Intern, **Spot & Tango**, New York, NY (Remote) — ***Jun 2025 – Aug 2025***

- Engineered scalable Airflow pipelines using Python, Singer, and GCP for automated data ingestion
- Parallelized Klaviyo API fetching, cutting ingestion turnaround time by ~96% for 135M+ event records
- Built a custom NetSuite RESTlet, reducing latency by ~97.2%; automated ETL across BigQuery, GCS, AWS S3

Data Engineer, **Atidiv**, Pune, India — ***Jul 2022 – Jul 2024***

- Built end-to-end ETL/ELT pipelines for startups using Python, SQL, Singer, and Meltano frameworks
- Cut Calendly Event Invitees Extractor turnaround time by 90% via parallelized processing
- Orchestrated pipeline deployment/monitoring across Databricks, Snowflake, and BigQuery

Graduate Research Assistant, **UVA Darden School of Business**, Charlottesville, VA — ***May 2025 – Aug 2025***

- Architected healthcare finance dashboard on GCP with serverless FastAPI backend, deployed via Cloud Run
- Built React + TypeScript SPA with US-state choropleth, KPI cards, and interactive filters; hosted on Firebase
- Optimized performance by pushing filters to SQL and caching client-side, cutting query latency by ~76%

Graduate Research Assistant, **University of Virginia**, Charlottesville, VA — ***Sep 2025 – Present***

- Tuned GPGPU-Sim parameters via LLM prompting; building RAG-based iterative search
- Reduced L2 cache metric error to near-0% via optimal few-shot LLM prompting

Graduate Teaching Assistant, **University of Virginia**, Charlottesville, VA — ***Sep 2025 – Dec 2025***

- Graduate TA for CS 4774 Machine Learning; built a Flask-based autoencoder grading platform
- Added token-auth submissions, async TorchScript evaluation, and SQLite leaderboard for robustness

Associate Software Engineer, **Informatica**, Bangalore, India — ***Dec 2019 – Jul 2021***

- Engineered and optimized ETL mappings/workflows for enterprise clients on Informatica platforms
- Resolved critical failures, delivered emergency fixes, and drove cross-team client POCs

TECHNICAL SKILLS

Languages: Python, C/C++, SQL
Frameworks: PyTorch, Hugging Face, TensorFlow, Keras, NumPy, pandas, Flask, Apache Kafka
Cloud and Data Management: AWS, GCP, Snowflake, Databricks
Tools and DevOps: Git, Apache Airflow, Rundeck, Docker, Kubernetes